

FSPP0034 PHYSICS II
Chapter 8 -10

Tutorial 5

1. A small candle is 35 cm from a concave mirror having a radius of curvature of 24 cm.

(a) What is the focal length of the mirror? 12 cm
 (b) Where will the image of the candle be located? $\frac{1}{12} = \frac{1}{35} + \frac{1}{d_i}, d_i = 18.26\text{ cm}$
 (c) Will the image be upright or inverted?
 inverted [Answer: (a) 12 cm (b) 76 cm (c) inverted]

2. An object 3.0 mm high is placed 18 cm from a convex mirror of radius of curvature 18 cm.

(a) Show by ray tracing that the image is virtual, and estimate the image distance.

(b) Using mirror equation, compute:

(i) image distance $\frac{1}{-9} = \frac{1}{18} + \frac{1}{d_i}, d_i = -6\text{ cm}$
 (ii) image size $m = \frac{-d_i}{d_o} = -\frac{-6}{18} = \frac{1}{3}$
 $m = \frac{1}{3} = \frac{h_i}{h_o} = \frac{h_i}{3\text{ mm}}, h_i = 1\text{ mm}$
 [Ans: (a) -6 cm (b) (i) -6 cm (ii) 1 mm]

3. The speed of light in ice is 2.29×10^8 m/s. What is the index of refraction of ice?

$n = \frac{c}{v} = \frac{3 \times 10^8}{2.29 \times 10^8} = 1.31$ [Ans: 1.31]

4. A flashlight beam strikes the surface of a pane of glass ($n = 1.56$) at a 63° angle to the normal. What is the angle of refraction?

$1.56 = \frac{\sin 63}{\sin r}, r = 34.83^\circ$ [Ans: 35°]

5. In searching the bottom of a pool at night, a watchman shines a narrow beam of light from his flashlight, 1.3 m above the water level, onto the surface of the water at a point 2.5 m from his foot at the edge of the pool (Fig. 1). Where does the spot of light hit the bottom of the pool, measured from the bottom of the wall beneath his foot, if the pool is 2.1 m deep.

$1.333 = \frac{\sin 62.53}{\sin r}, r = 41.73^\circ$
 $\tan \theta = \frac{0}{9}$
 $\tan \theta = \frac{1.3}{2.5}$
 $\theta = 27.47^\circ$
 $\theta = 62.53^\circ$
 $\tan 41.73 = \frac{x}{2.1}$
 $x = 1.87\text{ m}$
 $\text{total} = 1.87 + 2.5 = 4.37\text{ m}$
 [Ans: 4.4 m]

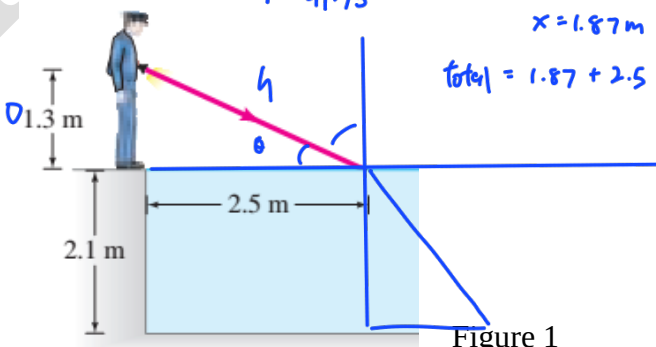


Figure 1

6. A sharp image is located 373 mm behind a 215-mm focal-length converging lens. Find the object distance (a) using a ray diagram, (b) by calculation.

$\frac{1}{21.5} = \frac{1}{d_o} + \frac{1}{37.3}, d_o = 50.75\text{ cm}$ [Ans: (b) 508 mm]

$$m = \frac{-d_i}{d_o} \quad d_i = -22.5 \text{ cm}$$

$$2.5 = \frac{-d_i}{d_o} \quad \frac{1}{p} = \frac{1}{q} + \frac{1}{-22.5}, \quad p = 15 \text{ cm}$$

- virtual image formed, upright, magnified

- convex lens is needed

$$-p = \frac{1}{0.15} = 6.7 \text{ D}$$

7. It is desired to magnify reading material by a factor of $2.5\times$ when a book is placed 9.0 cm behind a lens. (a) Draw a ray diagram and describe the type of image this would be. (b) What type of lens is needed? (c) What is the power of the lens in diopters?

[Ans: (a) virtual, upright, magnified (b) converging lens (c) 6.7 D]

8. Two 25.0-cm -focal-length converging lenses are placed 16.5 cm apart. An object is placed 35.0 cm in front of one lens. (a) Where will the final image formed by the second lens be located? (b) What is the total magnification?

$$(a) \frac{1}{25} = \frac{1}{35} + \frac{1}{d_i} \quad 67.5 - 16.5 \quad \frac{1}{25} = \frac{1}{-71} + \frac{1}{d_i} \quad (b) m = \frac{-d_i}{d_o} \times \frac{-d_i}{d_o} \quad \rightarrow = \frac{-87.5}{35} \times \frac{-18.5}{-71} = -0.651$$

[Ans: (a) 18.5 cm (b) $-0.651 \times$]

9. Both surfaces of a double convex lens have radii of 31.4 cm . If the focal length is 28.9 cm , what is the index of refraction of the lens material?

$$\frac{1}{f} = (n-1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right), \quad \frac{1}{28.9} = (n-1) \left(\frac{1}{31.4} + \frac{1}{31.4} \right), \quad 0.0346 = 0.0637n - 0.0637 \quad [Ans: 1.54]$$

$$0.0637n = 0.0983 \quad n = 1.543$$

10. The third-order bright fringe of 610 nm light is observed at an angle of 28° when the light falls on two narrow slits. How far apart are the slits?

$$d \sin \theta = m \lambda, \quad d = \frac{m \lambda}{\sin \theta} = \frac{3(610 \times 10^{-9})}{\sin 28^\circ} = 3.9 \times 10^{-6} \text{ m} \quad [Ans: $3.9 \times 10^{-6} \text{ m}$]$$

11. If 720-nm and 660-nm light passes through two slits 0.68 mm apart, how far apart are the second-order fringes for these two wavelengths on a screen 1.0 m away? [Ans: 0.2 mm]

$$y_m = \frac{m \lambda L}{d} = \frac{2(720 \times 10^{-9})(1)}{0.68 \times 10^{-3}} = 2.12 \times 10^{-3} \text{ m} \quad y_m = \frac{m \lambda L}{d} = \frac{2(660 \times 10^{-9})(1)}{0.68 \times 10^{-3}} = 1.94 \times 10^{-3} \text{ m} \quad \therefore 0.18 \times 10^{-3} \text{ m}$$

12. Two narrow slits separated by 1.0 mm are illuminated by 544 nm light. Find the distance between adjacent bright fringes on a screen 5.0 m from the slits. [Ans: $2.7 \times 10^{-3} \text{ m}$]

$$\Delta y = \frac{\lambda L}{d} = \frac{544 \times 10^{-9}(5)}{1 \times 10^{-3}} = 2.72 \times 10^{-3}$$

13. If 680-nm light falls on a slit 0.0365 mm wide, what is the angular width of the central diffraction peak?

$$\theta_w = 2\theta_1 = 2\sin^{-1} \left(\frac{\lambda}{a} \right) = 2\sin^{-1} \left(\frac{680 \times 10^{-9}}{0.0365 \times 10^{-3}} \right) = 2.13^\circ \quad [Ans: 2.13°]$$

14. Light of wavelength 580 nm falls on a slit that is $3.8 \times 10^{-3} \text{ mm}$ wide. Estimate how far the first brightest diffraction fringe is from the strong central maximum if the screen is 10.0 m away.

$$y_1 = \frac{(1+0.5)(580 \times 10^{-9})(10)}{3.8 \times 10^{-6}} = 2.29 \text{ m} \quad [Ans: 2.35 m]$$

15. When blue light of wavelength 440 nm falls on a single slit, the first dark bands on either side of center are separated by 55.0° . Determine the width of the slit. [Ans: 953 nm]

$$\theta_w = 2\theta_1 \quad a \sin \theta = n \lambda$$

$$55^\circ = 2\theta_1 \quad a = \frac{1(440 \times 10^{-9})}{\sin 27.5^\circ}$$

$$\theta_1 = 27.5^\circ \quad = 9.53 \times 10^{-7} \text{ m}$$